

Better growth starts here --- with you.♦

Coding Challenge – Lead AI Engineer

Thank you for your interest in joining our digital service line, *Elevate*, at Simon-Kucher & Partners.

At Simon-Kucher, we want to hire the smartest and most ambitious people, therefore we prepared a small coding challenge. It is important for us to see how you deal with this kind of task which is similar to some of the tasks that members of our teams work on every day.

All the data that you will need to use for this task are available via this link <https://filetransfer.simon-kucher.com/s/5w7HH7QYMBTri5N> (Password: #SK_AIEng_25!)

Please send back your solution via email to your HR contact once you have completed the challenge.

Background

Schröders GmbH is a consumer electronics company manufacturer and retailer operating primarily through digital B2C channels and retail B2B partnerships. Their pricing strategy has traditionally relied on competitor-based benchmarking, which worked well until recently. However, with the rise of new players post-pandemic and inconsistent pricing strategies across competitors, this approach has become unreliable—leading to revenue and margin losses.

To address this, *Schröders* partnered with us to define a more robust pricing strategy. Our assessment confirmed they have the necessary data, infrastructure, governance, and team capabilities to implement machine learning-based price optimization.

Project Overview

As part of an RFP, *Schröders* has asked us to develop an MVP price optimization model, packaged in an interactive Streamlit app, to demonstrate our approach and capabilities. They are convinced, we have the right team and set-up to deliver the MVP price optimization model, however, they are more curious about our ability to build an interactive interface as during the project the interface would be more complex and integrated across many back-end systems.

In addition to a dataset containing prices and sales, *Schröders* requests that we incorporate information from their Knowledge Database. For the MVP, they have provided three articles, with plans to extend this to the entire database in the future.

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Key Deliverables

1. Data Analysis & Model Building

- Develop a GenAI-driven pipeline to enhance the existing dataset with the three provided articles — 'Euro Electronics and Tech,' 'Headphones Today,' and 'Le Tech Quotidien.' The pipeline should automatically extract relevant information from the articles to enrich the dataset continuously.
- Analyze the enhanced dataset
- Build ML models to estimate price elasticity and optimize pricing

2. Interactive Visualizations via front-end app

- Commercial Impact Assessment
Use drop-downs and interactive chart visualizations to show the potential revenue/margin uplift under three scenarios:
 - Optimistic: price increases across all products
 - Conservative: price decreases across all products
 - Standard: 5% discount on top-selling products, increases on long-tail products
- Elasticity Visualization
Let users select a product from a dropdown and view its price elasticity curve
- Model Configuration
Allow users to:
 - Exclude (blacklist) specific products from training
 - Trigger model training directly from the app interface

To explain your solution - please develop an architecture diagram to explain your workflow.